

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We establish that physical scale and population size jointly govern the emergence of complex behavior in open-ended ecological environments. In our setting, agents are entirely unsupervised—furnished with neither explicit rewards nor learning objectives—and adapt over time solely through the coupled dynamics of reproduction, mutation, and selection. Because agents continually reshape both their environment and the composition of the surrounding population, the system constitutes a perpetually self-modifying ecology in which selective pressures are wholly endogenous rather than externally imposed. Rather than optimizing a single high-performance policy, we characterize how behaviors emerge, diversify, and stabilize across large populations governed by natural competition and environmental pressure. Leveraging modern hardware accelerators together with a purpose-built GPU-native multi-agent simulator, we scale both environment and population well beyond previously attempted sizes, sustaining populations that exceed 60,000 agents, each governed by its own independently evolved neural network policy. Across these simulations we identify a broad spectrum of emergent behaviors—long-range resource extraction, vision-based foraging, and predation among them—that arise directly and spontaneously from competitive and survival pressures, in the complete absence of any engineered objective. Through systematic manipulation of sensing modalities and environmental scale, we demonstrate that several of these behaviors emerge only once the environment and population are sufficiently large, and that increasing scale monotonically strengthens their stability and consistency. Set against a rich history of research in evolutionary settings, our scaling results on modern hardware chart a decisive and compelling direction for the study of ecology as an instrument of machine learning in an era of increasingly abundant computational resources and efficient simulation frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been studied across a broad range of biological disciplines, including evolutionary theory (Darwin, 1859; Simpson, 1944), ecology (MacArthur & Wilson, 2001), and genetics (Dobzhansky, 1937). Models of these emergent dynamics span theories of speciation and adaptive radiation (Coyne & Orr, 2004; Schluter, 2000) together with the mathematical analyses of population genetics (Fisher, 1930; Wright, 1931).

Within these fields, decades of mathematical modeling, laboratory experimentation, and field work have profoundly advanced our understanding of the mechanisms of evolution in nature. Studying emergent behavior directly in the natural world nonetheless remains fraught with fundamental obstacles: as ecosystems grow larger and more complex, they become correspondingly more difficult to control and to measure, and even where controlled manipulation of large-scale natural settings is feasible, it risks damaging or displacing the very wild populations under study. Simulation offers a principled route around these constraints, affording complete observability together with precise experimental control over conditions that are otherwise intractable to manipulate.

To surmount these obstacles and elucidate the emergence of complex behavior under competitive and environmental pressures, we study the open-ended evolution of neural network policies in large-

scale ecological simulations. In this setting, agents possess neither a specified objective nor a reward signal; they instead collect resources in order to survive and reproduce according to the intrinsic dynamics of the environment, and policy changes propagate solely through mutation as parents transmit their policies to their offspring. Capitalizing on the rapid growth in the computational resources available to researchers, we investigate the open-ended evolutionary dynamics of populations of these simple digital organisms at scales not previously attainable, thereby furnishing a new set of tools for characterizing how environmental scale and population size drive behavioral emergence.

To this end, we employ a new JAX-based simulation environment that supports rapid, GPU-native evaluation of large grid worlds. Our larger experiments comprise over 60,000 individual agents distributed across a physical area exceeding 1,000,000 grid cells—a regime that lies well beyond the scale of prior open-ended ecological simulations. Agents in this environment must navigate to locate food and other resources in order to survive, and may reproduce by accumulating resources sufficient to instantiate a mutated copy of themselves. A 3D visualization of the environment is provided in Figure 1.

Our aim is twofold: to illuminate ecological phenomena through simulation, and to establish ecology as a viable mechanism for producing machine intelligence. Recent years have made clear that in AI, scale does not merely improve models—it fundamentally transforms their capabilities (Wei et al., 2022a). Despite this insight, embodied learning has remained largely confined to comparatively small environments with modest populations. Our work therefore poses a distinct question: what capabilities emerge when we scale not a single model but an entire ecosystem? Just as reasoning emerges in individual models only beyond a certain size threshold (Wei et al., 2022b), we show that the strategies which evolve within populations likewise emerge only once environmental size and complexity attain sufficient richness.

Through an extensive suite of experiments, we identify multiple regimes in which sensor configuration and environmental scale decisively shape the behaviors that emerge from simulation and, in turn, precipitate long-term ecological consequences. Concretely, we demonstrate that agents equipped with simple compass sensors reliably evolve long-range resource-gathering expeditions—a behavior that emerges far less consistently at small scales—and that agents endowed with vision sensors evolve markedly more effective foraging and attacking behavior than agents lacking them; in both cases the effect strengthens systematically with scale. Taken together, these experiments establish that the scale and complexity afforded by modern hardware accelerators open a promising new avenue for the study of how evolution in ecological environments gives rise to rich, previously inaccessible behaviors.

2 RELATED WORK

The emergence of life has inspired computational research since the field’s inception. Conway’s Game of Life (Games, 1970) established that populations governed by simple local rules can produce intricate global structure. Early computer programs such as Tierra (Ray, 1992) and Avida (Brown, 1992) modeled life as self-replicating programs that evolve through mutation, while Reynolds (1987), in the same spirit as Conway, derived emergent flocking and schooling behavior from purely local rules of motion. These foundational works established that open-ended complexity can arise from minimal generative rules—a principle that our work carries to a substantially larger scale.

Researchers have applied these principles to evolve the locomotion morphology (Sims, 1994a, b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025) of virtual agents, while neuroevolution methods have evolved neural network architectures directly (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). Our work departs from these approaches in a fundamental respect: rather than engineering agents to adapt more effectively to their environment under a reward or objective signal, we characterize the behaviors that emerge spontaneously from ecological pressure alone, and we quantify precisely how environmental scale and complexity govern those behaviors.

Evolution has likewise inspired the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) recruit the principles of

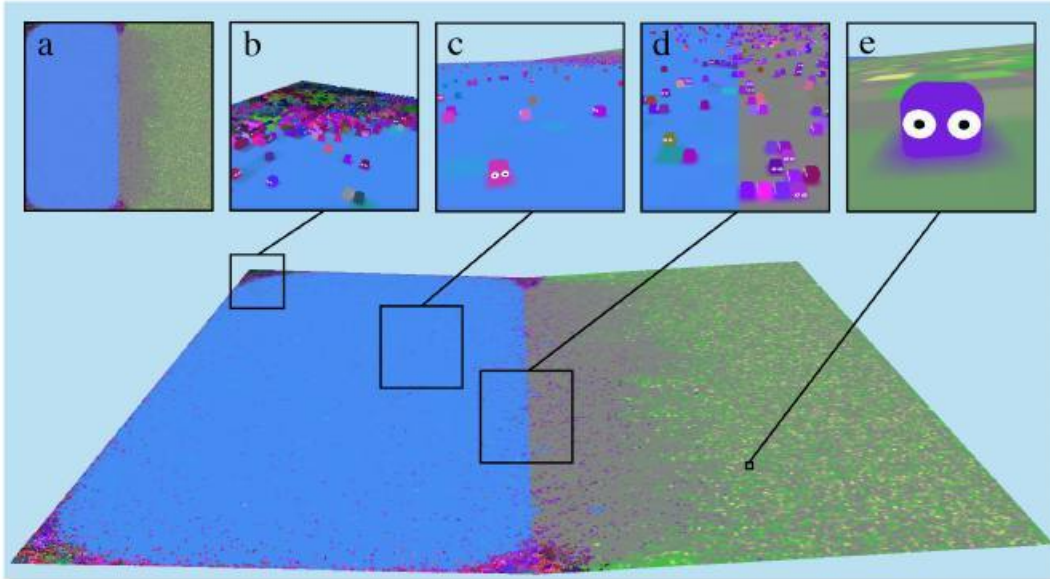


Figure 1: A 3D visualization of a 512×512 map in our environment. Inset (a) presents a top-down view of the entire map; (b) shows a group of agents clustered in a corner; (c) shows agents in the water; (d) shows agents travelling back and forth between water and land; and (e) shows a close-up of a single agent gathering resources.

evolution and natural selection for optimization, while related methods explicitly reward novelty among the resulting solutions (Lehman & Stanley, 2011; Cully et al., 2015). Such approaches have been used to optimize neural networks for tasks such as Atari games and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). Although our work likewise employs evolutionary mechanisms to update agents, its object is fundamentally distinct: we study emergent behavior itself, rather than developing a novel optimization algorithm or applying an existing one to a fixed objective.

A rich body of work likewise draws on biological inspiration to develop organisms within an environment. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Damer (1998), and Tisue et al. (2004). More recent efforts employ cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025), while Mordatch & Abbeel (2018) and Park et al. (2023) exploit simulated environments to investigate the emergence of language and communication between agents. The emergence of behavior through autocurricula (Leibo et al., 2019)—the principle that populations of agents constitute an increasingly challenging learning landscape for one another—has similarly been examined, though only in environments with populations substantially smaller than ours (Leibo et al., 2018; Team et al., 2021). Complementarily, Wang et al. (2019) and Aki et al. (2024) pursue the inverse strategy, evolving environments specifically optimized for agent learning; although these works evolve the environment rather than the agent, they directly motivate our approach by demonstrating that environmental complexity is a potent driver of sophisticated behavior.

Most closely related to our work are several efforts that simulate populations of agents interacting within complex environments. Baker et al. (2019) examine a Hide-and-Seek task in which a small number of hider agents learn to exploit environmental tools to evade seeker agents, both trained via reinforcement learning. Other works similarly train populations for specific tasks: playing video games (Jaderberg et al., 2019), manipulating objects (Petrenko et al., 2021), or capturing agents (Zheng et al., 2017; Yamada et al., 2020). Although many of these works explore emergence, they invariably do so under fixed tasks and externally specified objectives. The Neural MMO framework (Suarez et al., 2019; 2023) adopts a more open-ended posture, placing agents within a massively multi-agent game setting and optimizing only for survival rather than fixed objectives; its population, however, remains capped at a fixed number of agents (experiments examine up to 128 concurrent

agents) and lacks generational inheritance, which sharply limits the study of population dynamics. Hamon et al. (2023) aligns more closely with our setting through variable population sizes, genetically inspired evolution, and generational timelines, yet operates in a substantially smaller and less feature-rich environment and does not directly interrogate the role of scale. Similarly, Lu et al. (2024) simulates agents evolving to gather resources, but likewise operates at a far smaller scale (a maximum of 256 agents) and emphasizes tool use through programmable robots. Each of these efforts advances the study of emergent ecological dynamics; a comprehensive account, however, demands the simultaneous confluence of several ingredients that no prior system has combined: populations that grow and shrink naturally, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at scales where collective patterns can crystallize. Our work is the first to unify all of these elements within a single system, thereby rendering directly observable a class of ecological phenomena that remained inaccessible in prior work.

Finally, we build on recent work that emphasizes environmental speed and scale, employing optimized libraries to increase the computational throughput and the number of agents able to interact within game-like environments (Koyamada et al. 2023; Matthews et al. 2024; Lechner et al. 2023). We extend this trajectory of progress in computational efficiency to a new domain, adopting analogous JAX-based acceleration techniques to attain both large scale and fast simulation specifically within open-ended ecological environments.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize the interactions between populations of agents and their environment as a class of multiplayer game we term an **Ecological Game (EG)**. In the manner of Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG possesses underlying Markovian transition dynamics, yet is expressly constructed to represent time-varying populations of agents in an objective-free setting. As in an SG, an EG comprises a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ that maps a state s_t together with a per-player action set $a_{t,i}$ to a distribution over successor states s_{t+1} . The EG departs from the SG formulation in precisely two respects: it discards the reward function entirely, and it introduces a population function $\Gamma(s_t)$ that reports which agents in I are currently alive and identifies their parents, so that birth and death events can be inferred directly from the state. Just as MDPs and SGs admit partially observable counterparts—POMDPs and POSGs—an EG likewise admits a partially observable variant, which we denote a POEG. A POEG differs from an EG in that it precludes each agent from observing the entire state, instead incorporating an observation function $O(s_t, i)$ that maps a state and a player to that player’s local observation. Our environment is precisely such a POEG, its observation function determined by the sensor configuration detailed in the next subsection.

3.2 ENVIRONMENT

The specific POEG instantiated in our experiments is a large grid world populated with resources that agents must collect in order to survive and reproduce. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, in which s_t^m denotes the two-dimensional data associated with the map—the terrain and the spatial distribution of resources—and s_t^a denotes the per-agent data, comprising each agent’s position, health, and internal resources.

Terrain: Each map is defined by a rock layer that specifies the height of every grid cell. Water is introduced into cells whose rock value falls below a preset sea level and, over the course of an episode, flows downhill according to a discrete flow simulation driven by this height field. The experiments below employ six distinct global terrain configurations, shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. We conduct experiments across a broad range of grid sizes— 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 —scaling the initial population in proportion to the total grid area as the grid grows or shrinks. Once the simulation begins, the population fluctuates freely according to the intrinsic dynamics of the environment rather than under any externally imposed constraint.

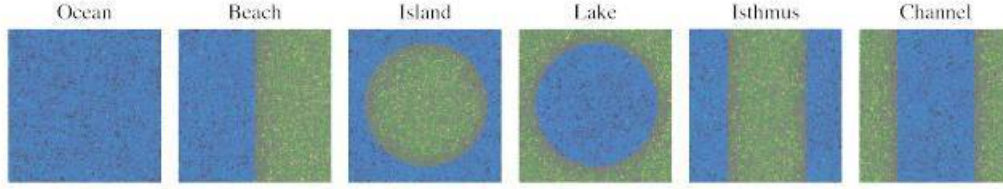


Figure 2: Overhead views of the six terrains used in our experiments, shown at the 256×256 grid size. Small dark spots denote individual agents, while green and yellow indicate concentrations of energy and biomass, respectively. Each image was captured 5000 steps into an episode—after most of the biomass in the water has been consumed, but before the agents have evolved to survive on land for extended periods—so that the characteristic early-episode resource distribution of each terrain is rendered visible.

Resources and Agent Health: To survive and reproduce, agents must collect three distinct resources—**water**, **energy**, and **biomass**. Biomass and energy are initially distributed at random throughout the environment but may thereafter be consumed and redistributed by individual agents.

Every action an agent takes expends a small quantity of energy and water. Spent water is returned to the landscape, where it flows downhill to the nearest body of water, whereas spent energy is consumed and destroyed. Biomass underwrites reproduction: because every agent requires a fixed quantity of biomass simply to exist, an agent cannot reproduce until it has accumulated surplus biomass sufficient to endow its offspring. Like water, total biomass is conserved across the simulation; unlike water, it neither flows downhill nor moves of its own accord. Energy, by contrast, is not conserved but instead accrues slowly on free-standing biomass through a photosynthesis process. These three coupled resource dynamics are deliberately designed to pose a strictly harder problem than the collection of a single food source, since they require agents to balance three interdependent resource budgets simultaneously. For ablations on the resource rules, see Appendix [F.2](#)

Each agent additionally possesses health points (HP) that it must maintain to avoid death. Over-spending a critical resource triggers rapid health deterioration, though an agent may expend energy and water to recover HP gradually over time. Agents may also attack one another, reducing the target’s HP, and each agent sustains a small but monotonically increasing amount of damage as a function of age, so that an agent eventually dies of old age if no other cause intervenes first. When an agent’s HP reaches zero and it dies, whatever resources it was carrying are deposited into the environment at the grid cell it occupied; consequently, an agent that attacks and kills another may subsequently consume the victim’s resources—a rudimentary yet fully emergent form of predation. For ablations on the HP hyperparameters, see Appendix [F.8](#)

Sensors: Agents perceive the environment through four sensors, each supplying a distinct modality of information. The **internal** sensor reports the agent’s own age, health, and carried resources; the **external** sensor reports the external resources present within the agent’s own grid cell; the **compass**, inspired by the magnetoreceptors found in birds and other animals, encodes the agent’s global heading as a 4-way one-hot vector; and **vision** supplies a 7×7 top-down rendered patch centered on the agent, comprising three color channels together with a fourth channel that encodes the local elevation gradient. For ablations on the vision configuration, see Appendix [F.6](#)

Actions: Agents select from a finite set of discrete actions organized into five categories. The first is **Rest**, which performs no operation and incurs the lowest energy cost. Three **Move** actions—rotate clockwise, rotate counterclockwise, and advance one grid cell—govern locomotion; because no two agents may occupy the same cell simultaneously, a move action is canceled whenever an agent attempts to enter a cell already held by another agent or whenever two agents contend for the same cell. A single **Attack** action kills every agent within the 3×3 square directly ahead of the actor. For ablations on attack strength, see Appendix [F.7](#) Three distinct **Eat** actions, one per resource type, each transfer a fixed quantity of the corresponding resource from the environment into the agent’s stomach—which has a fixed maximum capacity—whenever the agent occupies a cell containing that resource.

Finally, a single **Reproduce** action produces an offspring via single-parent reproduction. This action succeeds only when the agent has accumulated resources sufficient to endow its child, which is then

instantiated in the cell directly behind the parent. The child’s policy is a mutated copy of the parent’s weights, and the child immediately inherits a fixed fraction of the parent’s resources.

In the experiments that follow, we systematically disable specific sensors or the attack action so as to isolate the ecological effect of each individual agent capability. We consider three sensing combinations: internal+external resources only (**R**); internal+external+compass (**RC**); and internal+external+compass+vision (**RCV**). The suffix **+A** indicates that attacking is enabled; for example, (**RC+A**) denotes an agent equipped with resource sensors and a compass that is additionally able to attack.

3.3 POLICIES

Agent policies are realized as small, memoryless multilayer perceptrons (MLPs); we adopt memoryless agents by deliberate design choice, so as to simplify the computational model and minimize the parameter count. The raw values from each active sensor are flattened, normalized to the range $[-1, 1]$, and passed through separate linear encoding layers, each with output dimension 64. The resulting per-sensor embeddings are summed and propagated through a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer is linear with output dimension equal to the number of admissible actions, and actions are sampled from a softmax distribution whose temperature is itself an evolvable parameter, mutated alongside the network weights. Weights are stored in half-precision bfloat16 to support larger populations and faster run times. Depending on the set of active sensors, each agent’s network comprises approximately 10k–25k parameters. For ablations on network size and architecture, see Appendix F.5

In every run, the weights of the initial population are set by Kaiming initialization (He et al., 2015), with all bias vectors initialized to zero. The policy of an agent born mid-episode is generated by copying its parent’s weights and perturbing every mutable weight with additive Gaussian noise of standard deviation 3×10^{-2} , a mutation scheme consistent with prior neuroevolution research (Such et al., 2017). Although more elaborate mechanisms such as crossover are available, we deliberately restrict reproduction to this single, parsimonious mutation operator in order to isolate the effect of environmental scale in the simplest setting possible. Empirically, the low-precision floating-point representation necessitated a somewhat larger mutation rate; we ablate this choice in Appendix F.4

Each agent additionally carries a 3-channel trait that governs its rendered color on the map. This color trait mutates in the same manner as the network weights and determines the agent’s appearance to other agents equipped with the vision sensor, thereby enabling vision-equipped agents, in principle, to discriminate among distinct types of neighbors.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented end-to-end in JAX (Bradbury et al., 2018) and executed on Nvidia H100 and A40 GPUs within a purpose-built, GPU-native simulation environment. Unless otherwise noted, each run proceeds for 2M environment steps or until extinction, with per-seed wall-clock times spanning under an hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on a single H100 GPU. Appendix E details the simulator architecture, reports additional scaling measurements, and discusses the considerations governing reproducibility.

4 EXPERIMENTS

The experiments below establish the effects of sensing modality, environmental scale, and population size on the emergent behavior of agents. Across both of our principal experimental settings—long-distance resource gathering and visual foraging with predation—a single, consistent picture emerges: richer sensing and larger environmental scale each produce emergent behavior that is more sophisticated, more effective, and more stable.

4.1 LONG-DISTANCE RESOURCE GATHERING

Our first experiment investigates the emergence of an agent’s capacity to travel long distances inland in search of resources. We demonstrate that in environments with clear directional structure, agents

equipped with a compass (**RC**) consistently evolve substantially more effective long-range resource-gathering behavior than a baseline agent class (**R**) that lacks a compass, and that the emergence of this behavior becomes both more consistent and more effective as environmental scale increases.

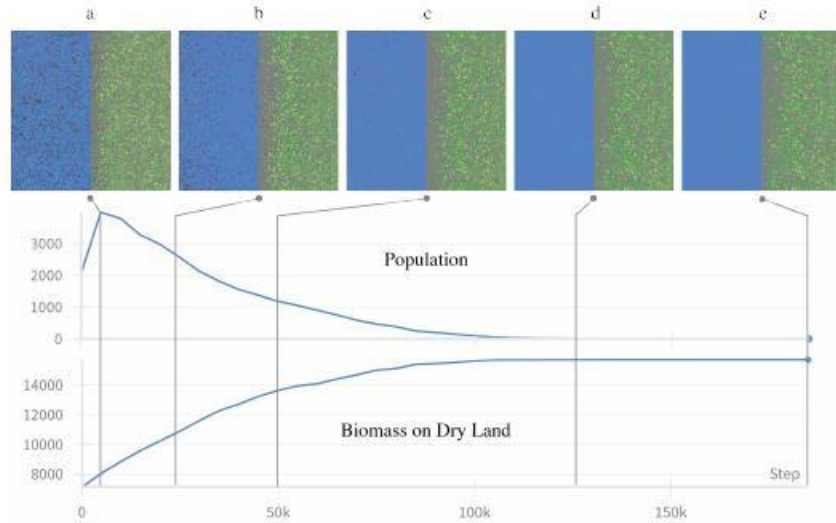


Figure 3: A representative trial with compass-free (**R**) agents in a 256×256 **Beach** environment. The top chart plots population over time; the bottom chart plots the free biomass on dry land. Absent a compass, agents fail to return from land to water, so that biomass steadily migrates onshore (bottom) and the population collapses to extinction (top).

We begin with a representative example of the compass-free (**R**) agent class, shown in Figure 3. By 5000 time steps (a), all initial agents on dry land have perished for want of water, while those in the water have multiplied rapidly on the strength of the rich initial resources; at this early stage, the population has not yet evolved policies capable of efficiently seeking out resources. From this early population peak, agents in the water occasionally wander onto land (b), where they frequently cannot navigate back to the ocean and consequently die, depositing their biomass on land. By 50,000 time steps (c), this gradual attrition has nearly doubled the uneaten biomass on land. The resulting transfer of resources from water to land contracts the population, since the water no longer retains biomass sufficient to sustain a large population. By 125,000 steps (d), almost all biomass has migrated onto land and the population is nearly extinct; the final agent dies around 185,000 steps (e).

This example lays bare the ecological consequences of the agents’ inability to return to water: the death of an individual not only removes it from the population but also reshapes the distribution of resources in ways that reverberate through the entire population. At the same time, this ecological catastrophe creates a decisive evolutionary opportunity for any agent that discovers how to walk onto land, forage in the rich biomass deposits left by its dead ancestors, and safely return to water—precisely the mining behavior we characterize next.

In the next example, shown in Figure 4, agents are additionally equipped with an internal compass (**RC**). Initially a similar pattern unfolds (a): the population contracts as free biomass on land rapidly accumulates. In this case, however, some agents exploit their compass to drift into the relative safety of the corners. Around 150,000 steps (b), agents begin to “mine” the biomass on land, using the compass to navigate back and forth between land and water; by 200,000 steps (c) these miners have cleared the entire beach. Thereafter the population undergoes only minor fluctuations—with agents forming clumps along the edges (d)—yet remains stable for the full 2,000,000 time steps (e). The compass thus converts an otherwise inevitable collapse into a self-sustaining ecology.

We quantify these observations through a comprehensive set of **Beach** experiments spanning grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , with initial populations of 32768, 8192, 2048, 512, and 128 agents, respectively. Each configuration is run with four random seeds; for each run we record whether mining behavior emerges—operationalized as a decline of at least

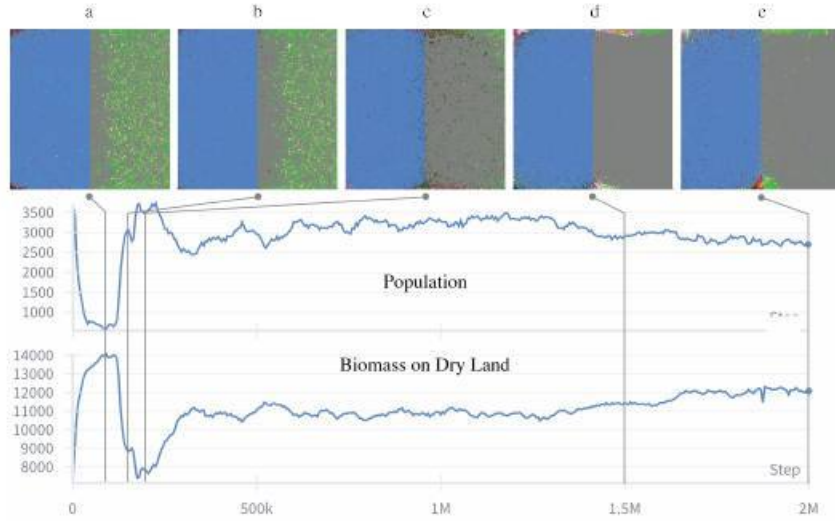


Figure 4: A representative trial with compass-equipped (**RC**) agents in a 256×256 **Beach** environment. The top chart plots population over time; the bottom chart plots the free biomass on dry land. In contrast to the compass-free case, agents evolve to mine the accumulated onshore biomass—sharply reducing free dry-land biomass (bottom)—and thereafter sustain a stable population for the full 2M steps (top).

Table 1: Mining events in the **Beach** world for populations of agents with a compass (**RC**) versus without a compass (**R**), across five world sizes and their proportionally scaled initial populations. denotes the number of seeds (out of four) exhibiting a mining event; denotes the number of seeds in which the population went extinct before 2M steps. Compass-equipped agents mine consistently whereas compass-free agents almost never do, and mining reliability rises monotonically with world size, reaching every seed at and above the 256×256 scale. This monotone dependence on scale, evident at a glance across the five columns, is the central pattern the table documents. See Table 3 in Appendix A for per-seed data.

	1024×1024		512×512		256×256		128×128		64×64	
	32768		8192		2048		512		128	
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

10% in the free biomass on dry land from its historical peak—together with the onset time of the mining event, determined by the local maximum in dry-land biomass, and the corresponding percentage drop in biomass. Table 1 summarizes the mining and extinction events, with per-seed detail provided in Table 3 in Appendix A. The comparison is unambiguous. Mining behavior emerges consistently in compass-equipped (**RC**) agents yet almost never in compass-free (**R**) agents; it emerges progressively more consistently as environments grow larger; and it arises in every single seed at and above the 256×256 scale. At smaller scales, by contrast, populations are markedly less stable and considerably more prone to early extinction. The compass and environmental scale therefore operate as two mutually reinforcing enablers of one and the same emergent capability.

To characterize these effects with greater statistical precision, we run additional seeds with (**R**) and (**RC**) agents at the 512×512 world size (32 seeds) and the 128×128 world size (64 seeds), and apply a Kaplan–Meier estimator to the probability of extinction and of mining as a function of simulator time. Extinction proves both rarer and slower to occur at the larger world sizes, whereas mining emerges both more frequently and earlier. Fitting a Cox proportional hazards model to the timing of mining discovery and of extinction across world sizes and compass usage corroborates these

findings quantitatively: larger world size substantially reduces the extinction rate and dramatically accelerates mining discovery relative to smaller worlds, and enabling the compass produces an effect of comparable magnitude. Full details appear in Appendix B

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for populations of agents with a compass (**RC**) versus without a compass (**R**) at the 512×512 grid size.  denotes the number of seeds (out of four) exhibiting a mining event;  denotes the number of seeds in which the population went extinct before 2M steps. Across all four terrains, the compass yields more frequent mining and fewer extinctions, establishing that the compass advantage generalizes well beyond the **Beach** setting to a diverse range of terrain geometries. See Table 4 in Appendix A for per-seed data.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

To assess how terrain configuration shapes the emergence of mining behavior, we conduct analogous experiments in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the 512×512 grid size with (**R**) and (**RC**) agents. Mining and extinction events for these experiments are summarized in Table 2, with per-run detail in Table 4 in Appendix A. Consistent with the **Beach** results, equipping agents with a compass yields earlier and more consistent emergence of mining behavior across every terrain, together with more stable populations and lower extinction rates. (**RC**) agents adapt more readily to the island and isthmus terrains than to the lake and channel, where they occasionally go extinct; we attribute this difference to terrain geometry, since in the island and isthmus a simple policy of walking due east or due west always returns an agent to water, whereas the channel and lake afford no comparably simple navigational rule. The compass-free (**R**) agent, by contrast, fails to adapt reliably to any of these terrains, faring least poorly on the island. The compass advantage therefore holds robustly across the full range of terrain geometries we examine. Figure 12 in Appendix D provides a visualization of mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We next conduct experiments in the **Ocean** environment to determine whether agents can adapt their vision sensors to locate free resources and prey on other agents effectively. We compare (**RCV+A**) agents—endowed with resource sensing, a compass, and a 7×7 visual sensor—against (**RC+A**) agents equipped with only resource and compass sensors, thereby isolating the marginal contribution of vision.

Figure 5 plots population, biomass utilization, movement actions, and eat actions for agents at the 512×512 and 128×128 grid sizes; plots for the remaining grid sizes appear in Figure 9 in Appendix C and summary statistics in Table 5 in Appendix A. Biomass utilization is defined as the percentage of the environment’s total biomass currently held within agents. In episodes where agents possess vision sensors (red lines), they spend less time moving and more time eating, sustain substantially larger populations, and achieve greater biomass utilization—direct and consistent evidence that these agents have adapted their visual sensing to forage more efficiently. The advantage conferred by vision, moreover, strengthens with scale: the variance between runs decreases substantially in larger environments, and the vision-equipped runs exhibit markedly more stable dynamics, with far less variation in these metrics over time.

Figure 6 reports the averaged attacks per agent and homicides per attack for these runs. Vision-equipped agents attack relatively infrequently yet sustain a high successful kill rate of approximately 75%, evidencing the emergence of a targeted, efficient predation strategy. In large grids, blind agents occasionally attain high kill rates as well, but only under far more unstable dynamics, and confounding factors complicate attributing this observation to any single cause. Visual inspection (see Figure 13 in Appendix D) reveals that these elevated kill rates typically coincide with clusters of agents accumulating in one location—clusters that almost never form in runs with visual agents—which

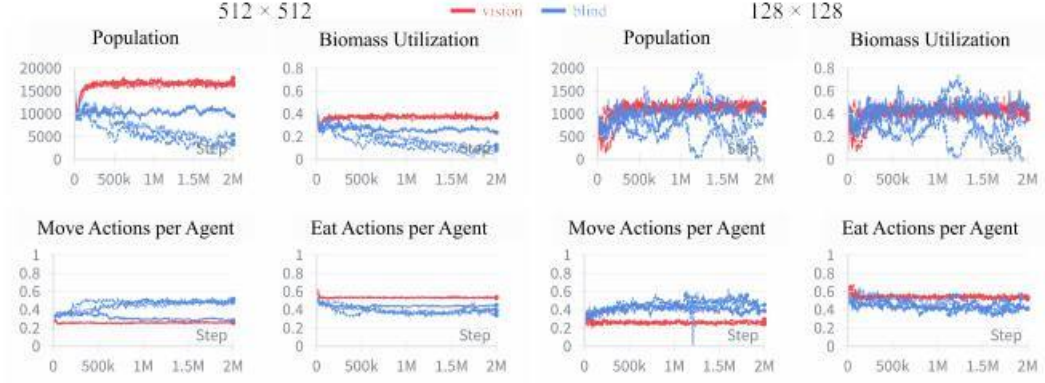


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agent populations are consistently larger and more stable and achieve greater biomass utilization—a direct consequence of reduced movement combined with more frequent and more effective eating. At larger scales these dynamics stabilize further, with markedly less variance across runs, demonstrating that larger environments amplify the foraging advantage that vision confers over blindness. See Figure 9 in Appendix C for the analogous comparison at grid sizes 1024×1024 and 256×256 .

further suggests that vision-equipped agents adapt to avoid one another. In smaller environments the difference between visual and blind agents is less pronounced, yet populations of visual agents remain considerably more stable throughout.

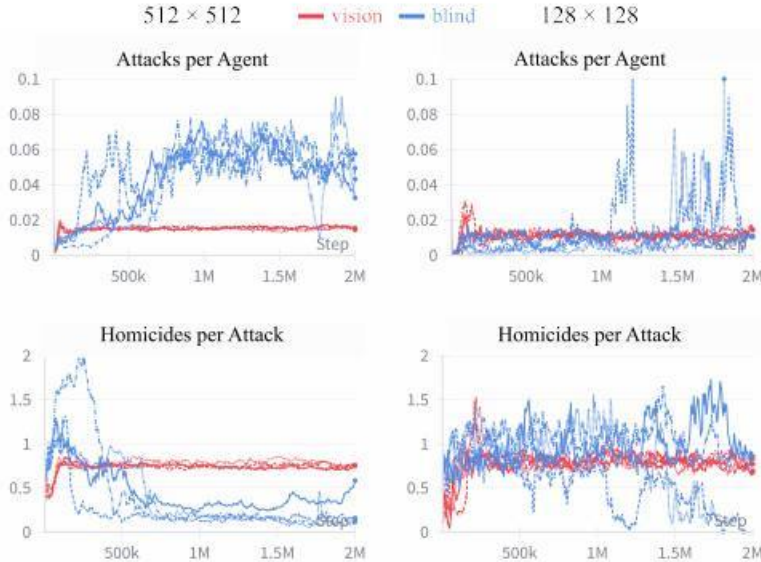


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less frequently but with substantially higher precision, evidencing the emergence of a specialized, targeted predation strategy rather than indiscriminate aggression. This dynamic grows more pronounced at larger scales. See Figure 10 in Appendix C for the analogous comparison at grid sizes 1024×1024 and 256×256 .

4.3 ABLATIONS

In Appendix F we systematically ablate the principal hyperparameters—initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8)—to characterize the sensitivity of the emergent behaviors to simulator configuration, and establish that our central findings remain robust across broad ranges of these settings. In Appendix F.3 we further demonstrate that the dynamics observed in larger environments cannot be attributed merely to self-averaging, confirming that environmental scale alters the underlying ecological dynamics rather than simply reducing stochastic noise.

5 CONCLUSION

Our study establishes open-ended ecological settings as a powerful and tractable paradigm for investigating the emergence of intelligence. Through controlled experiments in worlds of systematically varying scale and complexity, we demonstrate that environmental scale and sensory richness each contribute decisively to the development of sophisticated sensing and decision-making policies, and that the most complex behaviors we observe—long-range mining, visual foraging, and predation—emerge reliably only once scale is sufficient. Taken together, these results establish that ecological and evolutionary pressures, when coupled with environments that are sufficiently large and rich, constitute a natural pathway toward increasingly intelligent artificial agents. Our experiments further demonstrate that such rich behaviors can be evolved in populations of tens of thousands of agents within a modest hardware budget and without recourse to any explicit objective. We anticipate that these results will catalyze further exploration of ecology as a mechanism for both developing and understanding artificial intelligence.

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A ADDITIONAL DATA

Table 3: Per-seed mining events in the **Beach** world, contrasting agent populations equipped with a compass (**RC**) against a compass-free baseline (**R**) across five spatial scales.  indicates whether a run exhibited a mining event.  reports the percent decrease in free biomass on dry land over the course of the mining event, and  reports the approximate timestep at which free dry biomass attained the local maximum that marks the onset of the mining event; for runs exhibiting multiple mining events, we report the first.  reports the timestep at which the run went extinct, or n/a if it survived the full 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k 85.3 675			
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓ 10k 78. n/a			
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x 135			
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4 40k 81.7 3/4			
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x 45k			
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x 40			
2	x			n/a	x			1070k	x			165k	x			55k	x 25			
3	x			n/a	x			280k	x			125k	x			90k	x 40			
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4 4/4			

Table 4: Per-seed mining events across four distinct terrains—**Island**, **Lake**, **Isthmus**, and **Channel**—for **(RC)** agents versus **(R)** agents, establishing that the emergence of mining generalizes robustly beyond the **Beach** setting. Grid size is 512×512 with an initial population of 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging data for Vision **(RCV+A)** and Blind **(RC+A)** agents in the **Ocean** world across scales. denotes population size; denotes biomass utilization, the percentage of the system’s total biomass currently held within agents’ stomachs; denotes move actions per agent, the fraction of time an agent selected a move action, averaged across agents; and denotes the corresponding proportion of eat actions. Reported values of , , , and are averages of the metric over all steps within a seed. At every scale without exception, vision agents sustain larger populations and higher biomass utilization while moving less and eating more, quantifying the foraging advantage that vision confers.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation data for Vision (**RCV+A**) and Blind (**RC+A**) agents in the **Ocean** world across scales.  denotes attacks per agent—the fraction of time an agent selected the attack action, averaged across all agents—while  denotes homicides per attack, the number of agents killed per attack action. Reported values of  and  are averages over all steps within a seed. Vision agents attack far less frequently yet kill far more reliably per attack, precisely as expected of an efficient, targeted predation strategy. The value of 0.562 for  in seed 3 of the 128×128 (**RC+A**) run is an outlier: this seed went extinct before 2M steps, and the dwindling population inflates the attacks-per-agent metric.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We analyze the rates of extinction and mining emergence as a function of world size and of agents' access to global direction through the compass sensor. Extinction is defined as the population reaching zero individuals, and mining behavior is defined exactly as in the earlier experiments (a 10% drop in free biomass on land).

Runs are stratified jointly by world size and compass access. We simulated 64 runs each of 128×128 without a compass and 128×128 with a compass, together with 32 runs each of 512×512 without a compass and 512×512 with a compass. For each of the four resulting groups, we fit a non-parametric Kaplan–Meier estimator to the time-to-event distribution, treating runs that did not experience the event within 2M steps as right-censored. As Figures 7 and 8 demonstrate, worlds survive longer and produce mining sooner both when they are large and when agents are endowed with compass information, establishing that scale and sensing operate as independent and mutually reinforcing drivers of stable, adaptive population dynamics.

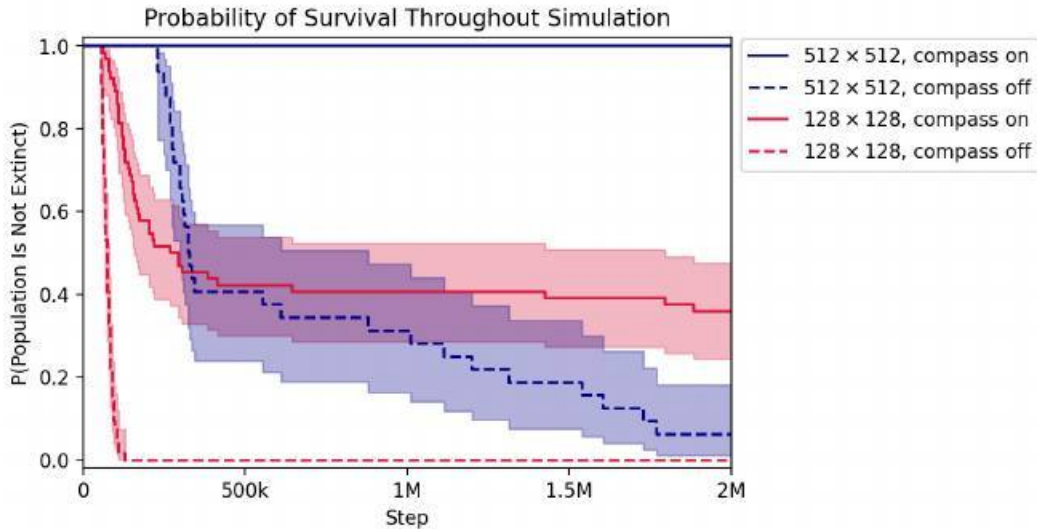


Figure 7: Probability that the population has not gone extinct, as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan–Meier estimator to each group of runs (e.g., 512×512 , compass on); shaded regions denote the 95% confidence interval for each curve. Each 512×512 group is fit on 32 runs and each 128×128 group on 64 runs. Larger worlds and compass access each shift the survival curves markedly upward.

We further quantify emergence and extinction timing with a Cox proportional hazards model—a semi-parametric survival model that relates covariates to event hazard rates (Table 7)—treating runs that experienced neither an emergence nor an extinction event within 2M time steps as censored. For extinction, both larger world size and compass use substantially reduce the hazard: the extinction rate in 128×128 worlds is roughly 25 times that of 512×512 worlds ($HR=0.04$), and enabling the compass produces an effect of comparable magnitude ($HR=0.03$). For mining emergence, larger worlds generate mining behavior far more rapidly ($HR=6.07$), and compass use accelerates emergence still more dramatically ($HR=34.59$). Both models discriminate strongly between groups (concordance 0.80 for extinction, 0.85 for mining), and every effect is statistically significant at $p < 0.005$. These results underwrite a coherent mechanistic account: larger worlds sustain larger populations, thereby reducing the probability of reaching the absorbing zero boundary while simultaneously exploring a larger phenotype space—by virtue of the greater aggregate volume of mutation—than smaller worlds.

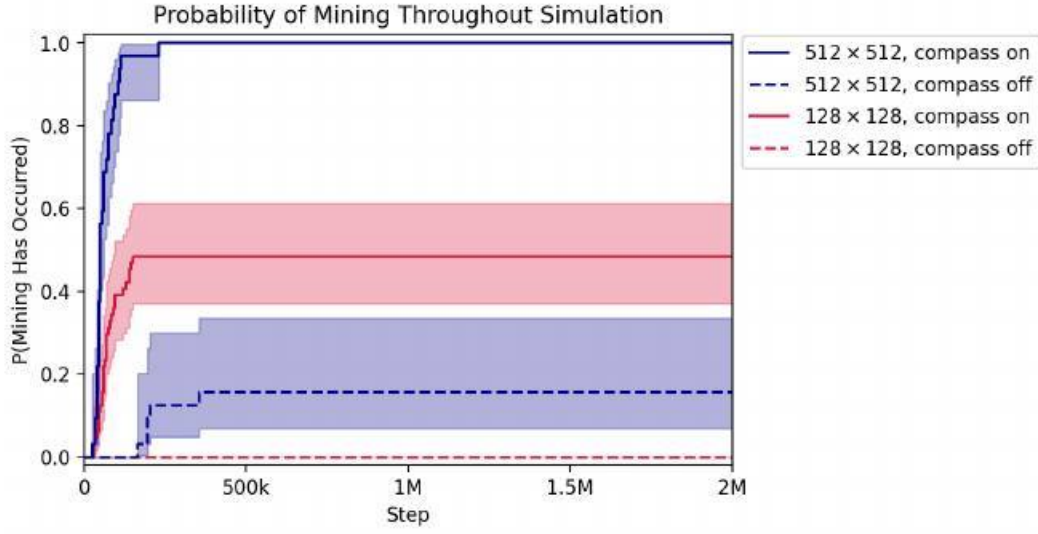


Figure 8: Probability that mining has evolved within the population, as a function of world size, compass access, and timestep. We fit a separate non-parametric Kaplan–Meier estimator to each group of runs (e.g., 512×512 , compass on); shaded regions denote the 95% confidence interval for each curve. Each 512×512 group is fit on 32 runs and each 128×128 group on 64 runs. Larger worlds and compass access each drive mining to emerge both sooner and in a greater fraction of runs.

Table 7: Cox proportional hazards models for extinction and emergence events, relative to a 128×128 , compass-off baseline. Hazard ratios (HR) below 1 indicate reduced hazard (a longer time to event); for example, the extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the rate observed in 128×128 worlds. Every reported effect is statistically significant ($p < 0.005$), and both models discriminate strongly between groups. The analysis comprises 192 runs in total: 32 seeds with compass on and 32 with compass off in the 512×512 world, and 64 seeds with compass on and 64 with compass off in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	<i>p</i> -value	HR	95% CI	<i>p</i> -value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

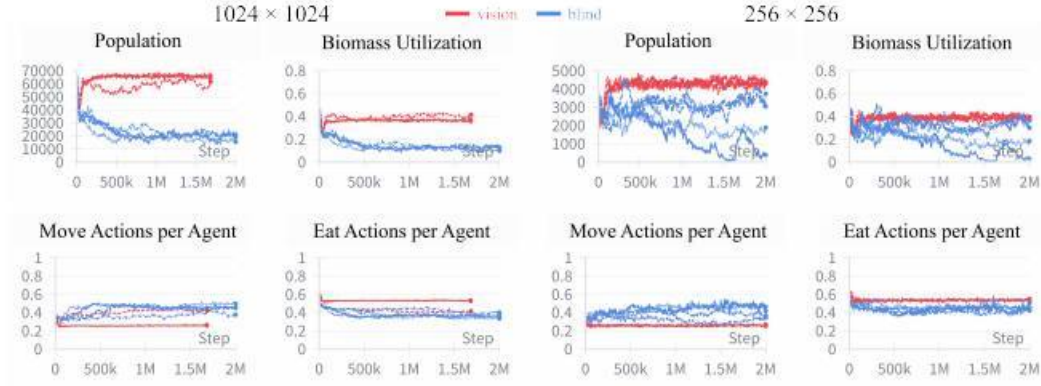


Figure 9: Foraging behaviors in the **Ocean** world at the 1024×1024 and 256×256 scales. Enabling vision (red) yields consistently more stable populations and greater biomass utilization, driven by more frequent and more effective eating together with reduced movement. At larger scales these dynamics stabilize further, exhibiting less variance across runs. Owing to limits on GPU hours, the vision seeds at the 1024×1024 scale ran for 1.7M rather than 2M steps.

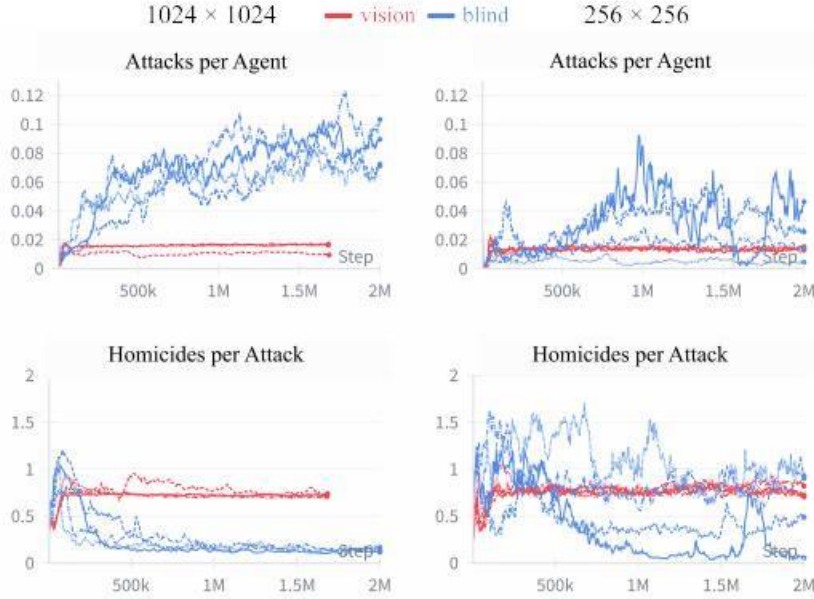


Figure 10: Predation behaviors in the **Ocean** world at the 1024×1024 and 256×256 scales. With vision (red), agents attack less frequently but with substantially higher precision, evidencing the emergence of a specialized, targeted predation strategy alongside the foraging capabilities documented in Figure 9. This dynamic is again more pronounced at larger scales, the blind agents (blue) exhibiting markedly greater variance at smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

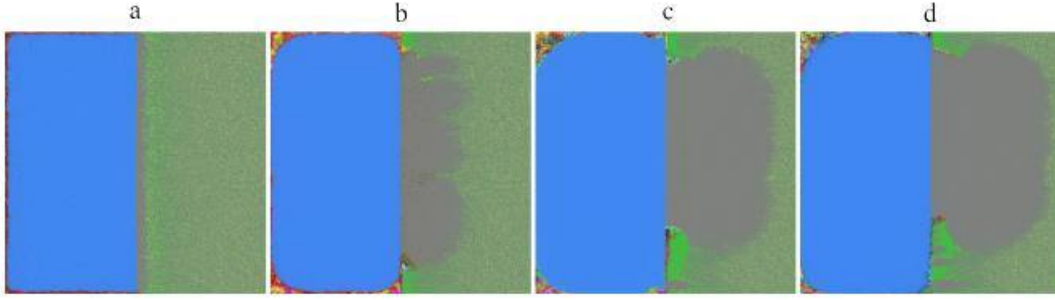


Figure 11: Larger-scale environments induce richer and more complex ecologies that give rise to distinctive spatial patterns. In this 1024×1024 **Beach** world, **(RC)** agents initially died at the shore (a), transferring biomass onto dry land; they subsequently evolved mining behavior and cleared biomass from the beach (b). As biomass re-accumulated at the ends of the beach (c), subpopulations independently re-evolved mining behavior, traveling inland along these secondary biomass deposits to extract the resources (d). This recurrence of mining across both space and time exemplifies the sustained adaptive dynamics that larger environments support.

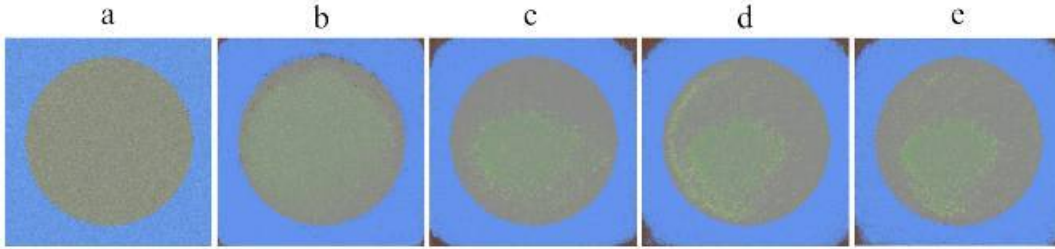


Figure 12: Mining behavior emerges robustly across varied terrains, not merely in the **Beach** setting. In this 512×512 **Island** world with **(RC)** agents, distinct regions of the shore are mined over the course of the run (b, c). Biomass is transferred back onto parts of the island once mining becomes difficult and agents die before reaching water (d); the agents then adapt to re-mine this accumulated wall of biomass, clearing it from the shore (e).

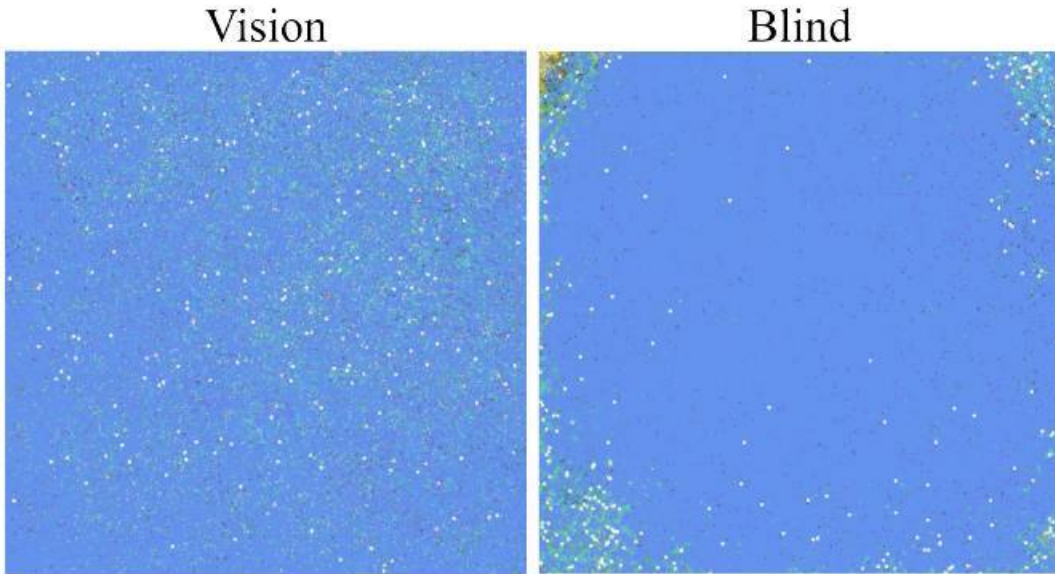


Figure 13: Representative runs in a 512×512 **Ocean** world with attacking enabled: vision **(RCV+A)** agents on the left, blind **(RC+A)** agents on the right. White squares mark locations where an agent is attacking. With vision, agents adapt to locate free resources throughout the ocean world and to prey on other agents far more effectively (see Figures 5 and 6). Blind agents, by contrast, explore the ocean for resources to a far lesser extent, instead adopting degenerate policies that cluster in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We provide details on simulator design, runtime, and experiment reproducibility.

Simulator design. All experiments are conducted with a JAX-based, GPU-accelerated simulator purpose-built for large-scale multi-agent ecosystems with open-ended populations. The simulator computes every component of the environment step on the GPU—terrain dynamics, resource flows, agent updates, and measurement alike—so that policy evaluation and simulation share a single JAX compilation pipeline, eliminating host-device transfer as a bottleneck. It exposes a deliberately minimal population-level interface: given the current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits, while the simulator handles births, deaths, and environmental transitions. This clean separation of concerns allows a single environment implementation to be reused, unchanged, across different evolutionary algorithms, reinforcement learning methods, and other population update rules. Because all operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, they fuse into a small number of GPU kernels that deliver the throughput required to reach the scales studied in this work; runtime measurements for a single H100 GPU are reported in Table 8.

A central feature of the simulator is its built-in tooling. A measurement API computes and records aggregate statistics at each step—biomass utilization, movement and eating rates, attack and homicide rates, and extensive environment-level parameters—which feed directly into our figures and tables, while an interactive 3D viewer renders terrain, water, resources, and agents from generated reports and supports inspection of individual agents (Figure 1). These tools proved indispensable for discovering and validating qualitative phenomena—long-range resource mining and predation patterns among them—that are not discernible from scalar metrics alone.

From a high-performance-computing standpoint, our simulator fills a distinct and previously unoccupied gap in the existing landscape. General-purpose agent-based frameworks such as MASON (Luke et al. 2005), FLAME-GPU (Richmond et al. 2024), and Griddly (Bamford et al. 2020) offer high configurability but demand substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study; GPU-native multi-agent RL environments, conversely, emphasize fixed agent counts and task-specific reward structures. By uniting GPU-native execution, explicit support for births and deaths with mutable heritable traits, and configurable resource and terrain dynamics within a single system, our simulator sustains open-ended populations of over 10^5 agents while remaining fully interoperable with diverse population algorithms. This combination constitutes a state-of-the-art platform for scaling ecological experiments such as those reported here, as well as a robust, reusable foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 reports runtime across grid sizes and maximum agent populations, with each run conducted on a single H100 GPU for a maximum of 2M steps. Although the table reports runtimes for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), runtime varies with agent network size, the set of active sensors, and the level of logging.

Our largest runs, at the 1024×1024 grid size, require only ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on a single H100, with peak memory usage of roughly 60GB—establishing that populations of tens of thousands of agents are tractable on modest, widely available hardware. **Ocean** runs are somewhat slower because both the agent attack action and vision sensing are enabled.

Reproducibility. Because our environments are large and randomized, the precise outcome of any given experiment can vary substantially from run to run, and we therefore report four distinct random seeds for each experiment. Owing to nondeterminism in certain low-level GPU operations, repeating an experiment with an identical random seed does not reliably reproduce the same trajectory: modern GPUs may execute parallel floating-point operations such as summations in differing orders across runs and may fuse operations in ways that alter rounding behavior, so that the minute discrepancies introduced by finite-precision arithmetic accumulate over many steps into appreciable trajectory drift. The use of 16-bit weights coarsens these rounding effects but is not their principal source. Crucially, notwithstanding this run-to-run variation at the level of individual trajectories, the

Table 8: Runtime in the **Beach** and **Ocean** worlds across grid sizes and maximum agent populations, quantifying the throughput that renders our largest-scale experiments feasible on a single GPU. Each (**RC**) agent carries a 22,465-parameter neural network and each (**RCV + A**) agent a 22,529-parameter network. We report simulation steps per second (hz) together with the total extrapolated wall-clock runtime in hours for 2M steps; effective runtime is lower for seeds whose populations go extinct before 2M steps.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	131072		32768		16384		8192		4096	
Environment	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

high-level patterns we report emerge robustly and reproducibly once the results of multiple seeds are aggregated.

F ABLATIONS

We present a series of ablations that probe the effects of specific hyperparameters and establish the robustness of our central results across a broad range of environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate how initial population size affects the consistency of mining behavior and population stability in a 256×256 **Beach** world, so as to disentangle the influence of initial population from that of environmental scale. Reported results are for (**RC**) agent populations; the compass-free baseline (**R**) exhibited a 0% mining rate and a 100% extinction rate at every initial population size except the largest tested, 8192, which yielded 25% mining and 75% extinction. We note that mining and extinction frequencies need not sum to 100%: a population may develop mining and subsequently go extinct, or may survive the full 2M steps without ever developing mining behavior.

As Figure 14 shows, initial population size exerts no appreciable effect on either mining consistency or extinction rate: populations seeded with only 128 agents evolve mining as frequently as those seeded with 8192 agents, and extinction rates remain essentially invariant across initial population size. This result is pivotal for the interpretation of our main findings, since it establishes that the scaling effects we report stem from environmental scale itself rather than from the initial number of agents.

Figure 15 clarifies the mechanism that underlies this near-invariance. Irrespective of its initial size, each run converges rapidly to a comparable population within the first 50,000 steps of simulation. This reveals that the environment implicitly defines a carrying capacity determined by the fixed resources in the system: populations initialized below the carrying capacity grow toward it, while populations initialized above it are driven down to it. Consequently, the effective population size is comparable across runs once the transient effect of the initial size has dissipated, leaving mining and extinction frequencies unaffected. Larger initial populations do, however, contribute modestly to the total resources in the system, since agents carry biomass that is deposited upon death; as a result, larger initial populations attain higher maximum populations on average over the course of simulation (Table 9), although this effect is too small to alter mining or extinction frequency in the present setting. Finally, although population size converges early, it does not remain constant thereafter: later in the simulation, populations across runs can diverge substantially, as the ecological consequences of agents evolving different abilities at different times reshape the subsequent resource distributions.

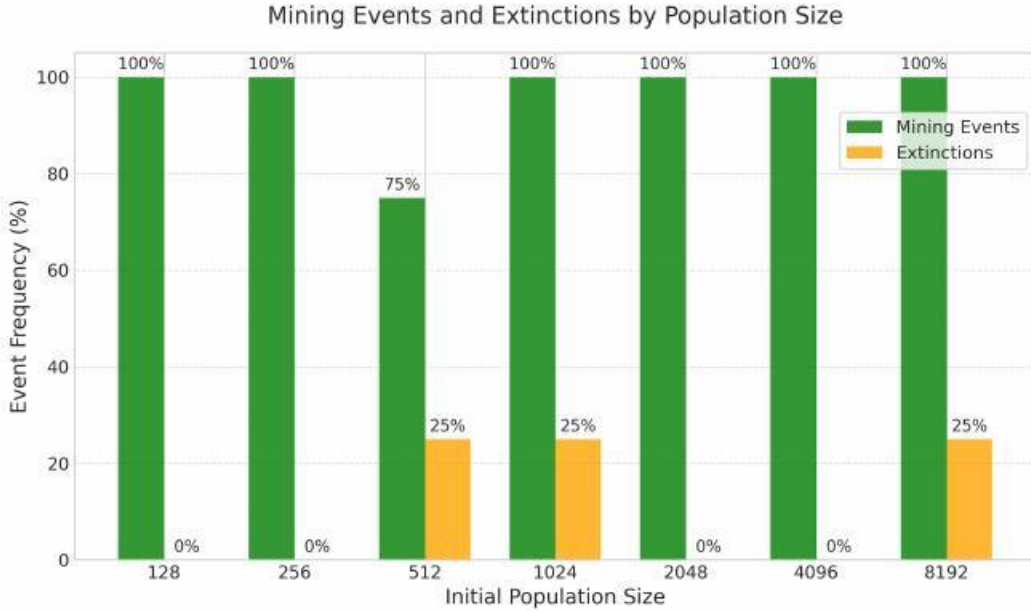


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world across initial sizes of (**RC**) agent populations. Frequencies are computed over four random seeds per initial population size; for example, a 25% extinction-event frequency means the population reached zero before 2M simulation steps in 1 of 4 seeds. Both frequencies remain essentially flat across initial population size, confirming that initial population is not the driver of the scaling effects reported in the main text.

Table 9: Maximum population size reached after step 0 for different starting population sizes, with means and standard deviations across four random seeds. Larger initial populations reach slightly higher maximum populations, with greater cross-seed variance, than smaller ones; the maximum population attained is nonetheless strikingly robust to the initial size: an initial population of 128—only 1.6% of an initial size of 8192—still attains 78.9% of the latter’s maximum population (3178 vs. 4026), underscoring the strength of the implicit carrying capacity.

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

F.2 VARYING RESOURCE RULES

We evaluate the effect of two resource hyperparameters—the biomass photosynthesis rate and the initial biomass site density—on mining and extinction frequency, in a setting otherwise identical to that of Appendix [F.1](#). Here the initial population size is fixed at 2048 and agents again possess a compass (**RC**).

The biomass photosynthesis rate governs the rate at which landscape biomass creates energy via photosynthesis, thereby replenishing resources for agents over time, while the initial biomass site density determines the fraction of grid cells that contain biomass at the start of simulation. All

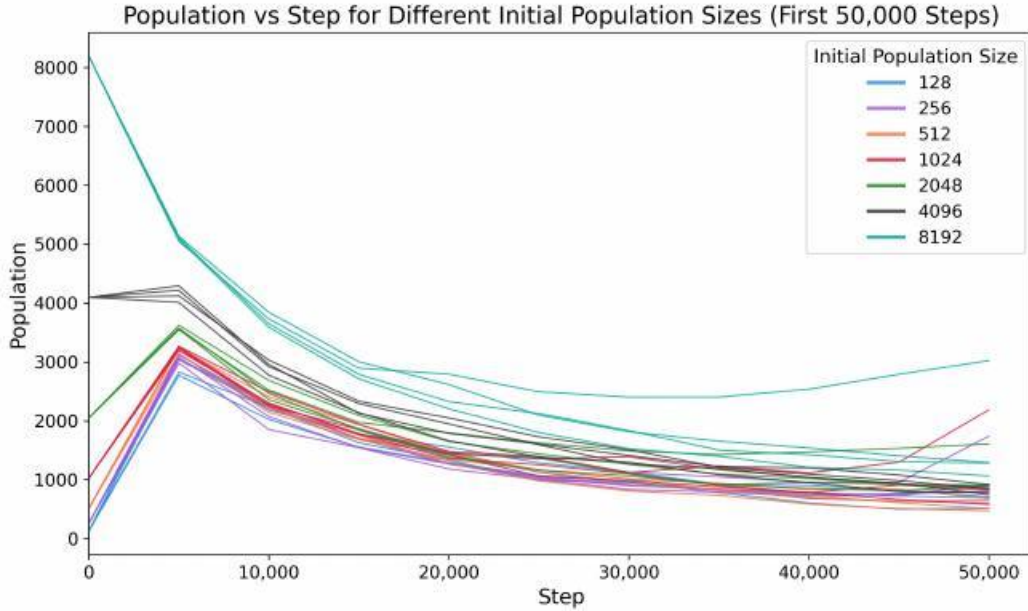


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with (RC) agents, with four seeds per initial population size. The environment sustains an implicit carrying capacity set by the resources available in the system, and every initial population size converges to this value early in the run, irrespective of where it began. Larger initial populations add modestly to the total resources in the system through the biomass deposited on agent death, thereby slightly raising the carrying capacity.

experiments in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of $1/4$.

Whereas varying the initial population size left mining and extinction consistency unchanged (Figure 14), varying the resource rules exerts a pronounced and systematic effect, as Figures 16 and 17 demonstrate.

The mechanism is intuitive and fully coherent with the carrying-capacity account above: larger biomass photosynthesis rates or initial biomass site densities directly raise the effective carrying capacity of the environment by increasing the resources available in the system, so that larger populations are sustained throughout the run, which in turn increases the likelihood of mutations favoring mining behavior while diminishing the chance of extinction. Our data bear this out quantitatively: averaging over the surviving seeds (omitting those that went extinct), runs with initial biomass site density $1/16$ reached a maximum population of 1608 after step 0, rising monotonically to mean maxima of 2466 at density $1/8$, 3786 at density $1/4$, and 5353 at density $1/2$.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

A natural concern is that the increased stability of mining behavior in large environments (Table 1) might reflect mere self-averaging rather than qualitatively different dynamics. By self-averaging we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, so that a single large system behaves like the average of many smaller systems of the same total size. On this view, a 512×512 **Beach** world would merely act as many loosely coupled 128×128 worlds run in parallel, and our scaling results could in principle be reproduced simply by running enough small worlds and averaging over random seeds. We now demonstrate decisively that this is not the case.

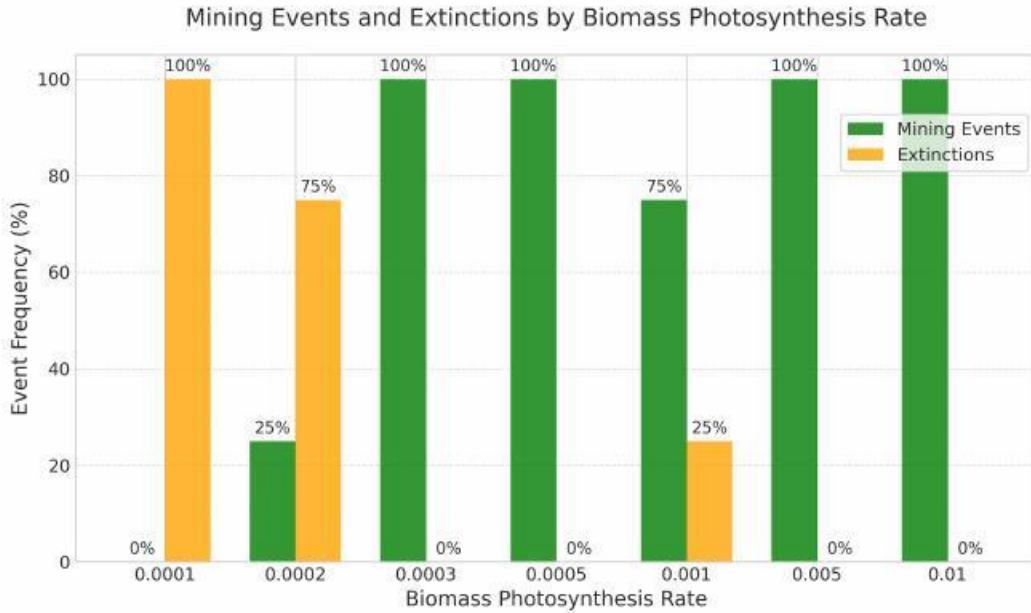


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the biomass photosynthesis rate. Frequencies are computed over four random seeds per setting. Mining emerges consistently at biomass photosynthesis rates of 0.0003 and above, tracking the concomitant increase in the environment’s effective carrying capacity.

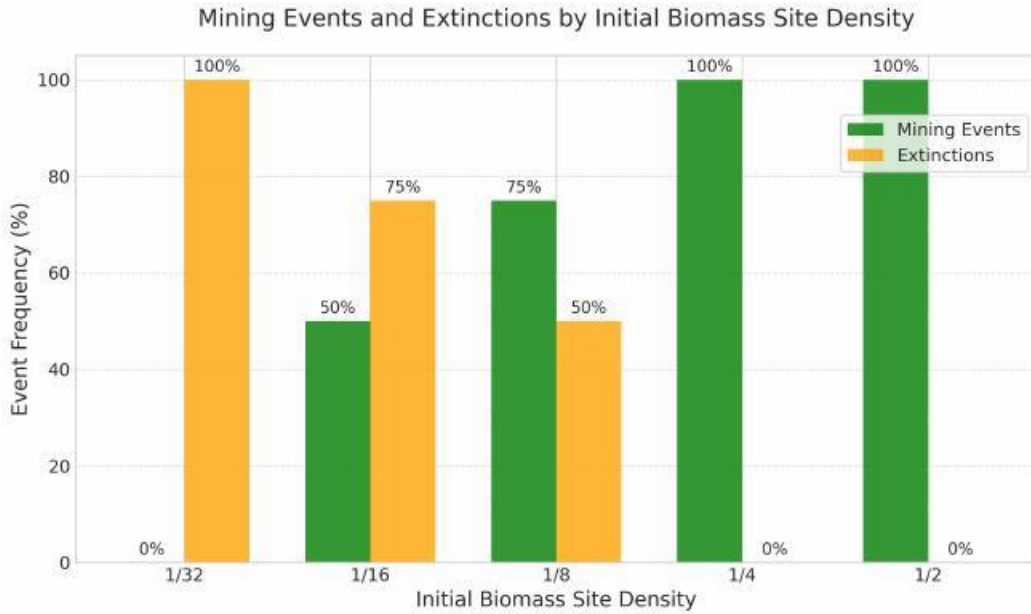


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the initial biomass site density. Frequencies are computed over four random seeds per setting. Higher initial biomass site densities yield more consistent discovery of mining and a lower chance of extinction, closely mirroring the effect of a higher photosynthesis rate.

To test this hypothesis, we hold the total “sample size” approximately fixed while varying how it is partitioned across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps with the same hyperparameters as in Section 4.1 and, as a control, we repeat the protocol for agents without a compass (**R**). Table 10 summarizes the frequency of long-distance mining and extinction in each configuration.

Were large environments merely self-averages of many smaller ones, mining and extinction rates would be roughly invariant to how grid cells and agents are distributed across runs, and 4 large worlds would exhibit essentially the same probability of discovering mining as 64 small ones. Instead, we observe a strong and unambiguous dependence on world size. For (**RC**) agents, mining emerges in every run with no extinctions at 512×512 , remains common but no longer guaranteed at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before reaching 2M steps. The (**R**) agents, by contrast, never develop mining and go extinct at every scale, confirming that the compass remains an indispensable enabler of long-range behavior even when total resources are held fixed. The control thus isolates world size as the operative variable.

Table 10: Mining and extinction events under a controlled trade-off between world size and the number of independent runs, with total grid cells and total initial agents held fixed across conditions. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , and 128×128 with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each size, the same total number of grid cells and initial agents is allocated across all runs by running 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged (🔍) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass (**RC**) and without (**R**). The strong dependence on world size, despite the deliberately matched totals, is flatly incompatible with a self-averaging explanation.

	512×512		256×256		128×128	
	8192		2048		512	
	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results are flatly incompatible with a self-averaging explanation. On the contrary, larger environments demonstrably enable and stabilize mining behavior in ways that cannot be reproduced by averaging over any number of small simulations.

Even with the total number of cells, agents, and simulation steps matched across conditions, concentrating resources into a few large worlds yields a regime in which mining is consistent and extinctions are rare, whereas dispersing the same resources across many small worlds yields populations that frequently fail to discover mining and are far more prone to collapse. Increasing environmental scale therefore does not merely smooth out stochastic noise; it qualitatively transforms the underlying ecological dynamics—for instance, by supporting larger, more spatially distributed populations and more persistent diversity among evolving policies. This finding decisively substantiates our central claim that scale is a genuine driver of emergent behavior rather than a statistical artifact.

F.4 MUTATION RATE SWEEP

We test the sensitivity of mining emergence to the mutation rate hyperparameter, which sets the scale of the Gaussian noise added to a child agent’s policy at birth and thereby determines how much a child differs from its parent. We again run experiments in a 256×256 **Beach** world with an initial population of 2048 (**RC**) agents for a maximum of 2M steps.

Figure 18 shows that across the entire interval between mutation rates of 6×10^{-3} and 7.875×10^{-2} , mining emerges in at least 3 of the 4 seeds, establishing that our Section 4.1 results at mutation rate 3×10^{-2} are robust throughout a broad range spanning more than an order of magnitude. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, culminating in extinction, whereas mutation rates that are too large (e.g., 3×10^{-1}) destabilize the population, consistently producing extinction with no mining adaptation. Our chosen rate thus sits comfortably within a wide window of viable values.

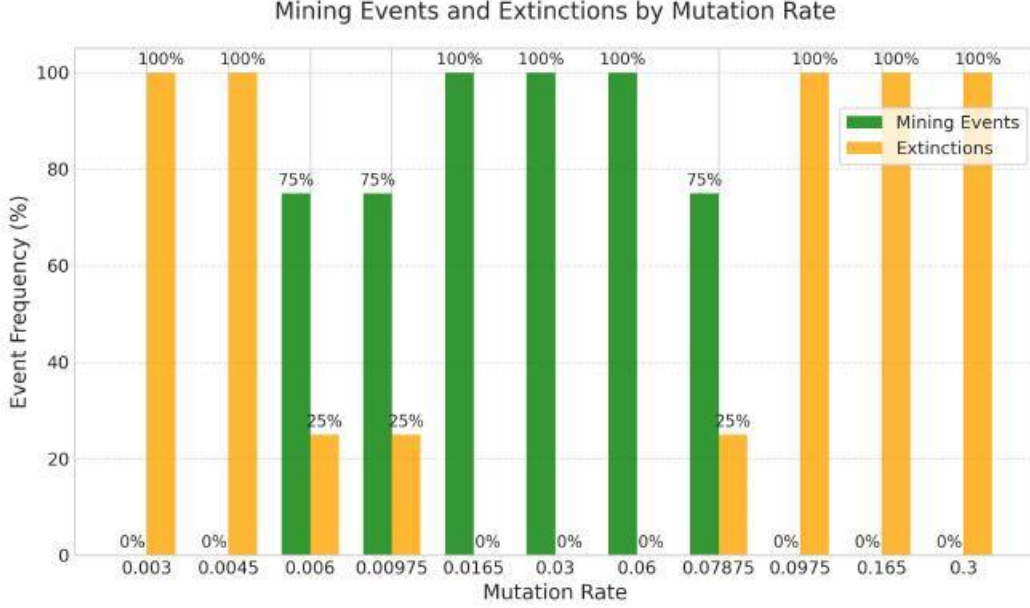


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world across mutation rates. Frequencies are computed over four random seeds per setting.

F.5 NETWORK SIZE

We study how the size and architecture of the agent policy network affect mining and extinction frequency, and how environmental scale modulates this effect. We investigate 256×256 and 512×512 **Beach** worlds with initial populations of 2048 and 8192 (**RC**) agents, respectively (scaling four-fold with the number of grid cells), and run (**R**) baselines for each experiment, which again yield rare-to-nonexistent mining and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, comprising 22465 parameters per agent (Table 11). Figure 19 shows that in the 256×256 **Beach**, holding the number of layers fixed while halving the channels to 32 (9185 total parameters) or doubling them to 128 (61313 parameters) exerts minimal-to-no effect on mining and extinction frequency. Network depth, by contrast, exerts a pronounced effect at this scale: every population of agents with just 1 layer and 32 channels went extinct, as did 50% of populations with 3 layers and 128 channels.

Crucially, scaling the environment to the 512×512 grid size produced markedly different dynamics for networks of varying depth (Figure 20). In the larger world, agent populations with 3-layer, 128-channel networks consistently mined and survived through 2M steps, and, strikingly, populations with the small 1-layer, 32-channel networks mined consistently as well—both configurations that had failed reliably at the smaller scale. We hypothesize that both smaller and larger networks struggle to discover new behavior, albeit for distinct reasons: a smaller network’s reduced capacity renders good solutions hard to find, whereas a larger network’s expanded search space renders promising new behavior slower to locate. Whatever the underlying cause, the fact that these networks fail so often in small environments yet succeed in larger ones furnishes further, independent evidence of the stability conferred by increasing scale. A larger environment sustains larger populations, thereby affording greater exploration of policy space; simultaneously, locally successful

mutants propagate more slowly across a large environment, which promotes diversity and thereby again ensures greater exploration. For further discussion of these effects, see (Mitchell et al., 2006).

Table 11: Total parameter counts of (RC) and (R) agent policies across the network architectures examined in this ablation. Even the largest configuration remains compact, underscoring that the emergent behaviors we study arise from remarkably small policies indeed.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

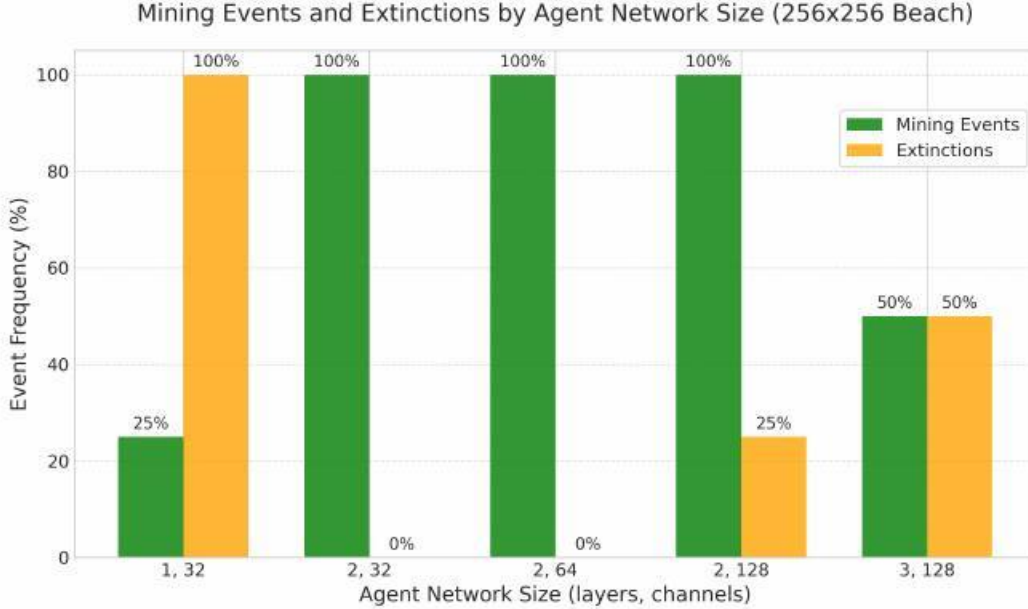


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting. At this smaller scale, both the shallowest and the deepest networks frequently collapse, whereas network width exerts little effect.

F.6 VARYING VISION RANGE

We assess how vision range and viewing configuration affect predation and foraging behaviors in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and 9×9 , and for each range compare the two viewing configurations shown in Figure 21 (a) Around, in which agents perceive grid cells in all directions, and (b) Front, in which agents cannot perceive the cells behind them. The two configurations hold the total number of visible cells equal, thereby isolating the effect of their spatial arrangement. The main vision experiments in Section 4.2 used the 7×7 Around setting.

These results demonstrate that a small 3×3 vision range yields behavior largely indistinguishable from no vision at all (Blind), whereas the 9×9 range performs comparably to 7×7 —clear evidence of diminishing returns to larger vision ranges under the current environmental configuration. This pattern admits a natural explanation: because the agents are memoryless, visual features at the

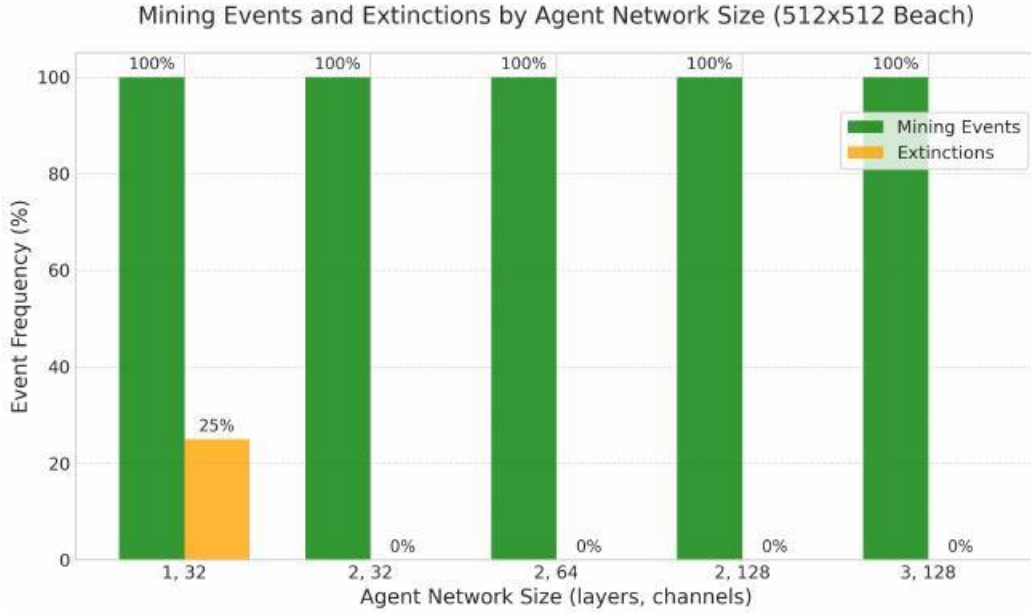


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting. In contrast to the smaller 256×256 case, the larger environment renders mining robust across network depths, encompassing even the shallowest and deepest configurations, and thereby demonstrating the robustness that scale confers to architectural choices.

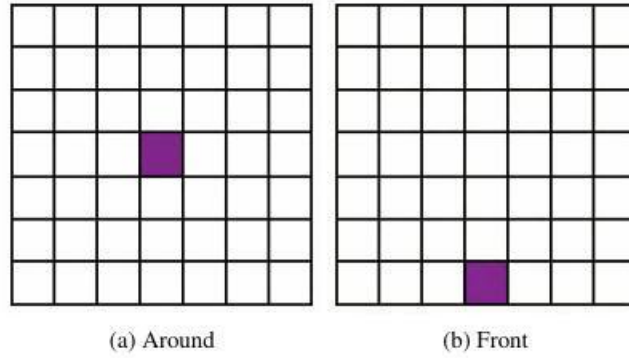


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell denotes the agent, oriented toward the top of the page. Both configurations expose the agent to the same number of cells, differing solely in whether that coverage is symmetric or concentrated ahead of the agent.

periphery of a 9×9 window are unlikely to alter the optimal action relative to a 7×7 window consistently enough to furnish a strong learning signal.

The data further indicate that at the 7×7 and 9×9 vision ranges, the Around configuration is slightly more favorable for population stability and for the effectiveness of the attack and eat actions, delivering higher average population 🧑, biomass utilization ⚙️, and homicides per attack 🎯 than the Front configuration. We again attribute this advantage to diminishing returns from perceiving more distant cells: in the Around configuration, a greater share of the visible cells lie nearby and therefore tend to carry more information relevant to near-term action than the distant cells visible under the Front configuration. At the 3×3 vision range, however, the trend reverses and the Front configuration becomes more favorable than Around, indicating that in this environment the marginal

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) across vision ranges and viewing configurations. See Tables 5 and 6 for the definitions of the metrics denoted by each icon. For each seed, every metric is averaged over all steps from step 500k onward, allowing the metrics to stabilize; reported values are the mean across the four seeds together with the standard deviation of the four seed averages. The 3×3 range performs comparably to Blind, whereas the 9×9 range matches the 7×7 range, evidencing clear diminishing returns to larger vision ranges.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
9×9	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

benefit of seeing one additional cell ahead under 3×3 Front outweighs that of seeing one cell behind under 3×3 Around—whose behavior, as noted above, most closely resembles the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study how reducing agent attack strength affects emergent behavior and population dynamics in the 256×256 **Ocean** world. We evaluate three attack strengths: (1) Full Attack, dealing 10 HP of damage and instantly killing any agent within the attack radius; (2) Half Attack, dealing 5 HP; and (3) Quarter Attack, dealing 2.5 HP. The main **Ocean** experiments in Section 4.2 used the Full Attack setting.

Reducing attack strength from Full to Half markedly diminishes both the effectiveness and the frequency of attacks: attacks per agent and homicides per attack both fall considerably (Figure 22). In consequence, populations under lower attack strength grow larger, with agents moving and eating more frequently and thereby attaining greater biomass utilization. Reducing strength further to the Quarter Attack setting yields diminishing returns, whether in the reduction of violence or in the facilitation of additional population growth. The underlying mechanism is a compounding one: when each attack deals less damage, a kill requires more attacks, which lowers the lethality per attack; but as lethality falls, attacking becomes less useful and hence less likely to be selected for, so that agents at lower attack strengths never become as skillful at attacking and instead reallocate their behavior toward foraging.

F.8 VARYING AGENT HP

We investigate the effect of reducing child agents’ starting HP and lowering the healing rate, which imposes pressure on agents to ‘grow’ over their lifetime in order to survive. The healing rate parameter determines how much HP an agent regenerates per healing step by converting available energy into HP; agents can therefore influence their own healing only indirectly, by maintaining sufficient energy.

We study two settings: (1) Full HP, in which child HP starts at 5 and the healing rate is 1 HP per 0.1 energy; and (2) Low HP, in which child HP starts at 1 and the healing rate is 0.5. Maximum agent HP is 10 in all settings, and the main experiments (Section 4) used the Full HP setting.

Across the two settings, homicides per attack, eat-action frequency, and biomass utilization exhibit no significant difference (Figure 23), demonstrating that the core foraging and predation behaviors are robust to this growth pressure. Under Full HP, populations grow slightly larger; under Low HP, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must concentrate on foraging early in life in order to grow to full size, which would account for this modest difference.

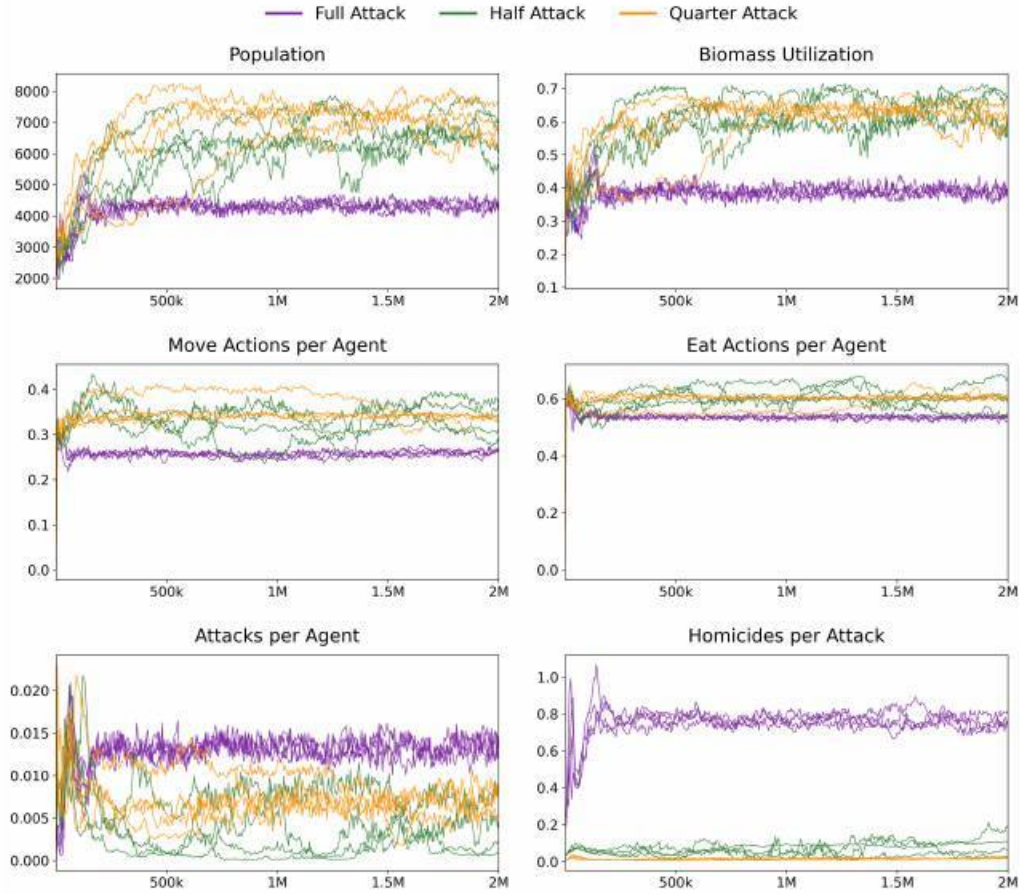


Figure 22: Population and action-frequency data over 2M steps in a 256×256 **Ocean** world for **(RCV+A)** agents across the Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted per setting. Weaker attacks yield less frequent and less lethal violence and, in turn, larger populations that forage more actively.

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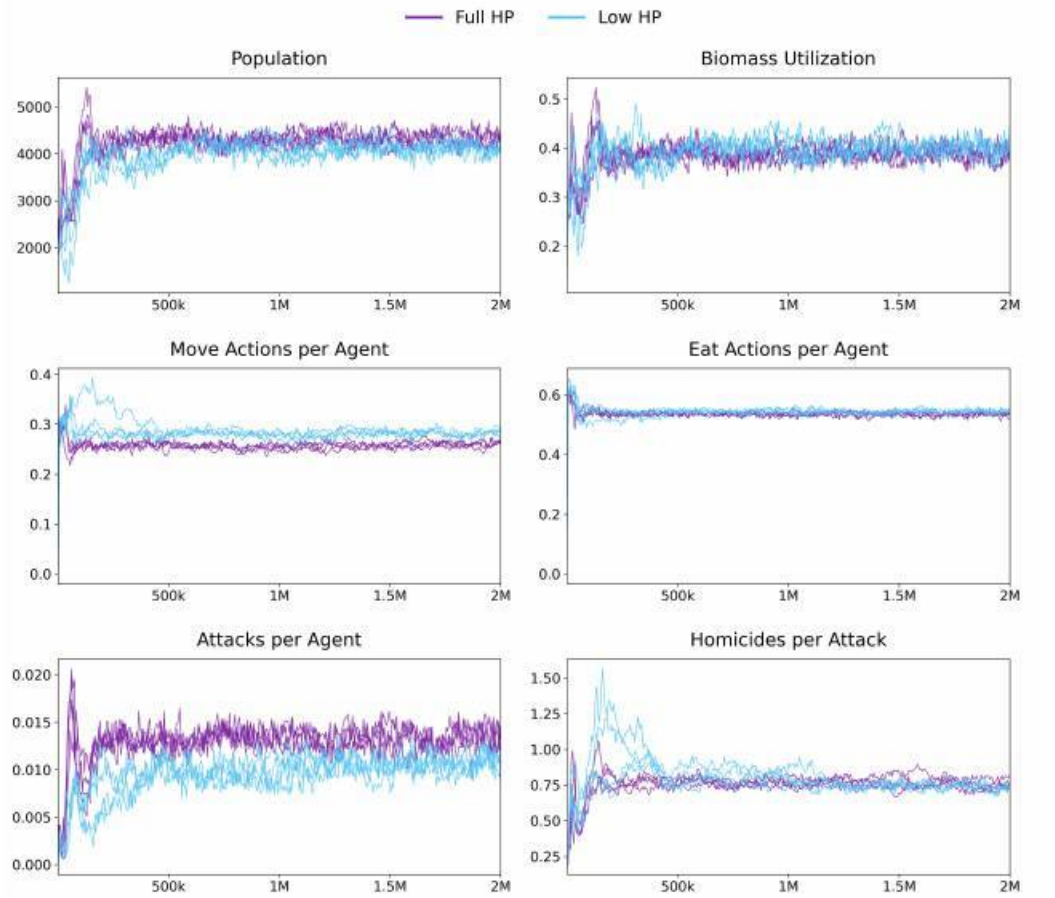


Figure 23: Population and action-frequency data over 2M steps in the 256x256 **Ocean** world for **(RCV+A)** agents across the Full HP and Low HP settings. Four seeds are plotted per setting. The core foraging and predation metrics remain largely unaffected by the HP setting, differing only modestly in population size and movement.